



Music Studio — Project Report

Technology: MERN Stack (MongoDB, Express.js, React.js, Node.js)

Deployment: Vercel (Frontend) · Render (Backend) · Cloudinary (Media)

April 2026

1. Introduction

Rama Records Music Studio is a full-stack web application for a professional music recording studio. It enables the studio to showcase songs, beats, services, and gallery, while allowing visitors to book sessions and contact the studio. A secure admin dashboard provides complete content management.

Problem Statement

Independent music studios lack a professional digital presence to showcase work, manage bookings, and engage clients efficiently.

Solution

A modern, responsive MERN-stack web application serving as a one-stop platform for visitors (browse, book, contact) and admins (manage content, track bookings, upload media).

2. Objectives

- Develop a responsive, visually appealing music studio website
- Implement secure admin panel with Google OAuth authentication
- Enable online booking and contact form with rate limiting
- Provide an in-browser music player for streaming songs and beats
- Use cloud-based media storage via Cloudinary
- Deploy using modern cloud platforms (Vercel + Render)

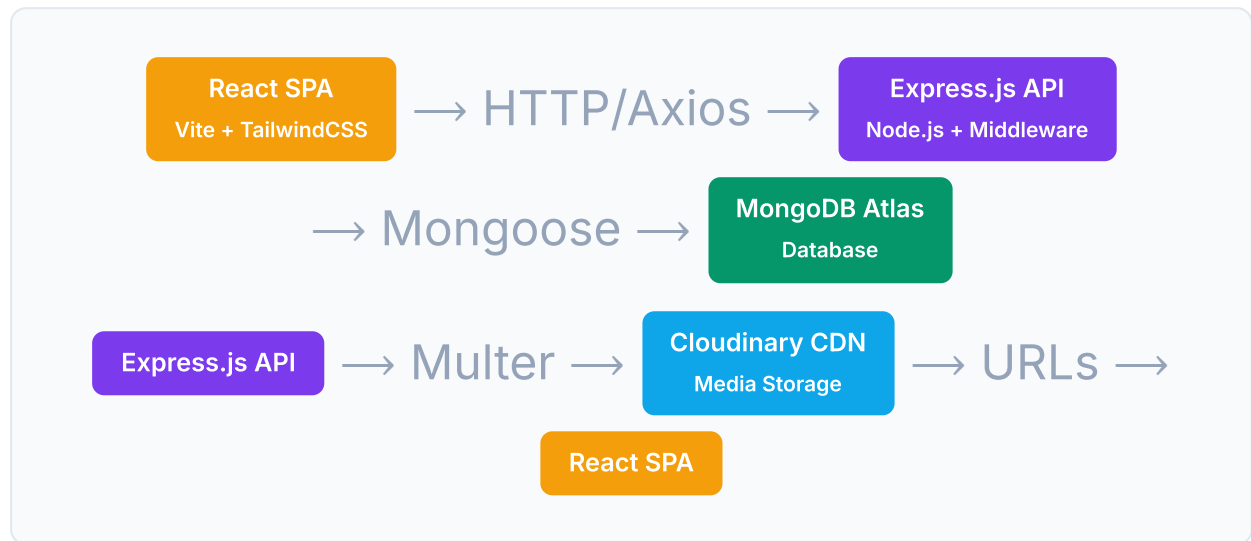
3. Technology Stack

Layer	Technology	Purpose
Frontend	React 19 + Vite 8	UI framework & build tool
Styling	TailwindCSS 3	Utility-first CSS
Animations	Framer Motion 12	Page transitions & animations
Routing	React Router DOM 7	Client-side navigation
HTTP Client	Axios	API communication
Backend	Node.js + Express 5	REST API server
Database	MongoDB + Mongoose 9	NoSQL document DB & ODM
Auth	JWT + Google OAuth	Token-based auth & SSO
Security	Helmet + express-rate-limit	HTTP headers & rate limiting
File Upload	Multer + Cloudinary	Media upload & CDN storage

Passwords	bcryptjs	Secure password hashing
Hosting	Vercel + Render	Frontend & Backend deployment

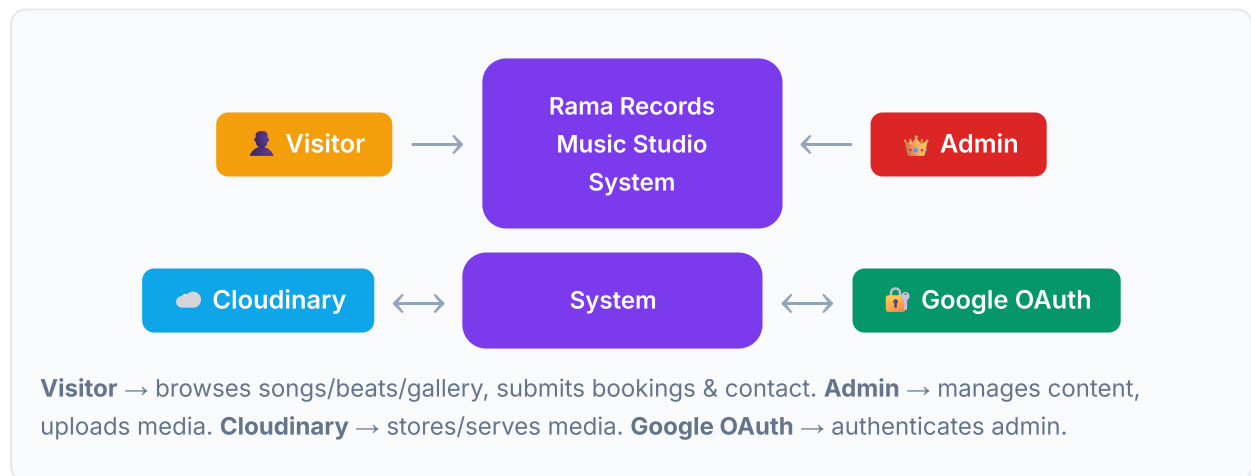
4. System Architecture

The application follows a **3-Tier Client-Server Architecture**:



5. Data Flow Diagram — Context Level (Level 0)

Shows the system boundary and external entities interacting with the application.



6. DFD — Level 1

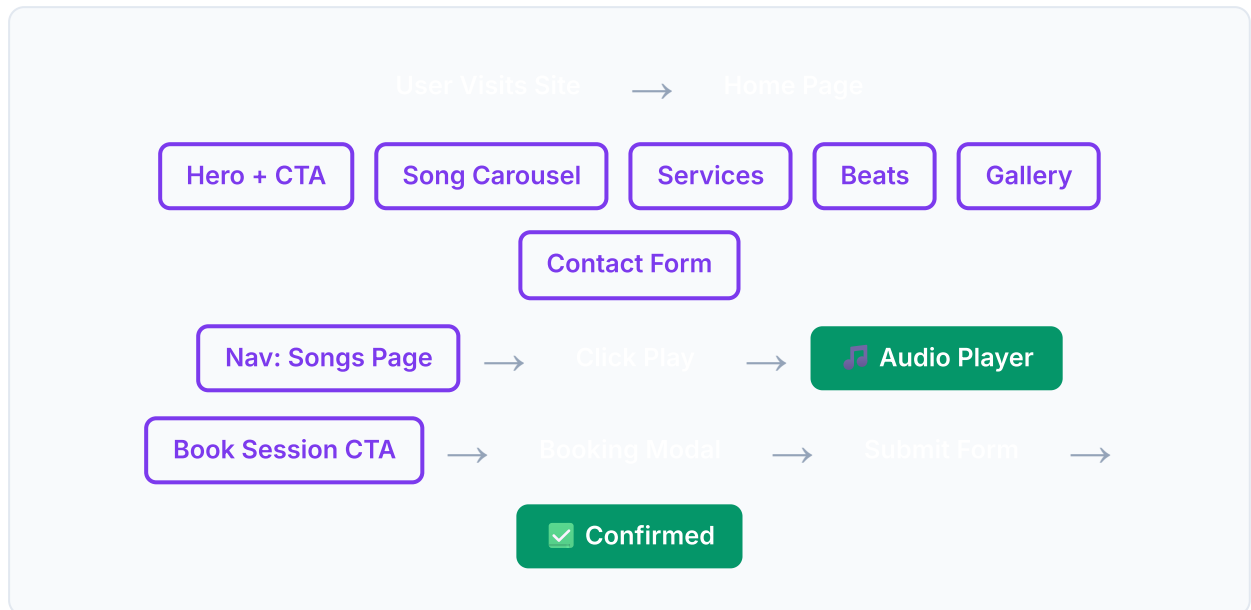
The system broken into its 6 major processes:

Process	Input	Output	Data Store
1.0 Authentication	Login credentials / Google token	JWT token	Users collection
2.0 Content Management	Song/Beat/Gallery/Service data	CRUD confirmation	Songs, Beats, Gallery, Services

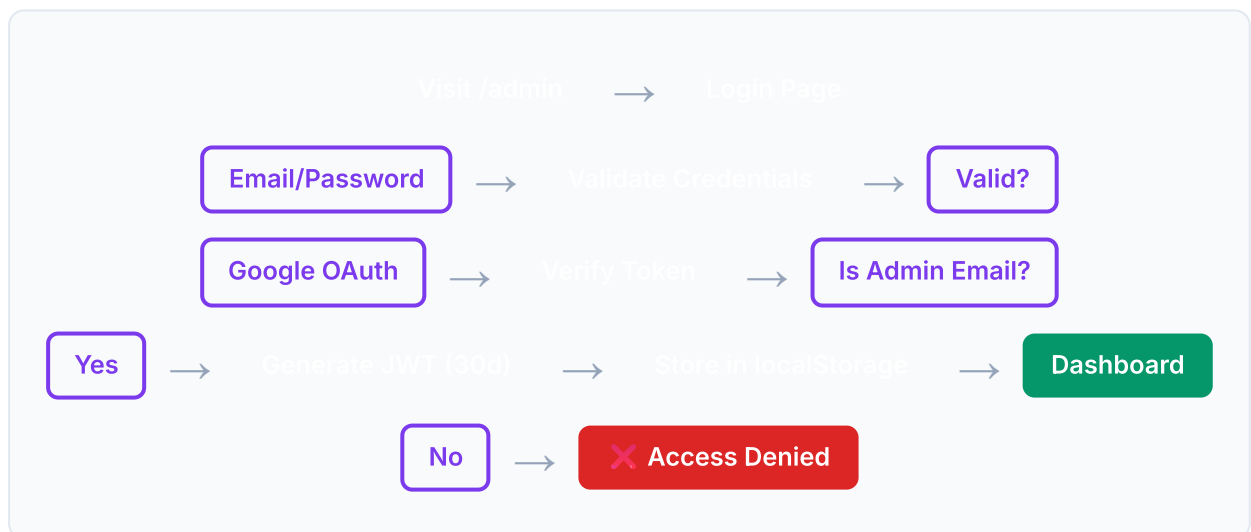
3.0 Booking Management	Booking form data	Booking status	Bookings collection
4.0 Contact Management	Contact form data	Message confirmation	Contacts collection
5.0 Media Upload	Audio/Image files	Cloudinary CDN URLs	Cloudinary cloud
6.0 Content Delivery	User browse request	Songs, Beats, Gallery, Services	All collections

7. Flowcharts

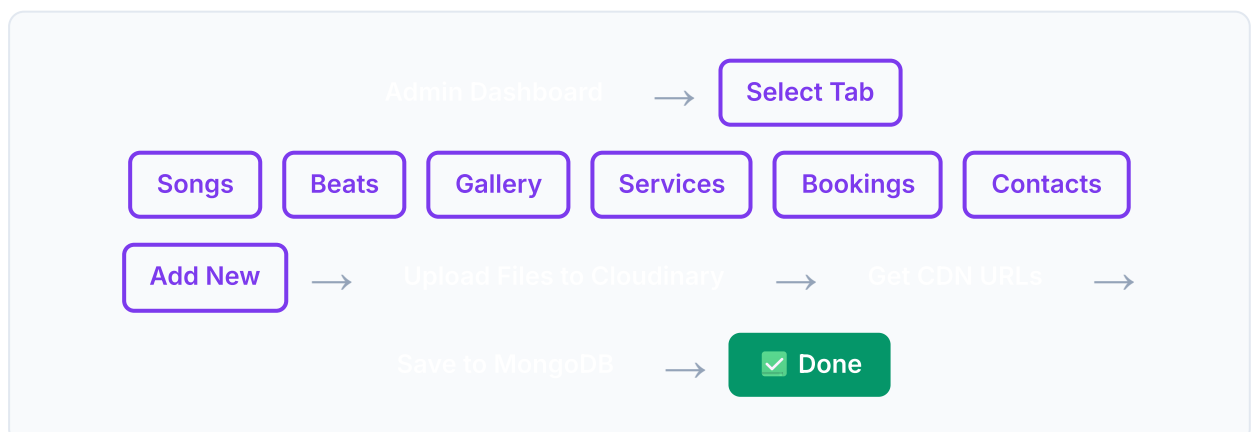
7.1 User Navigation Flow



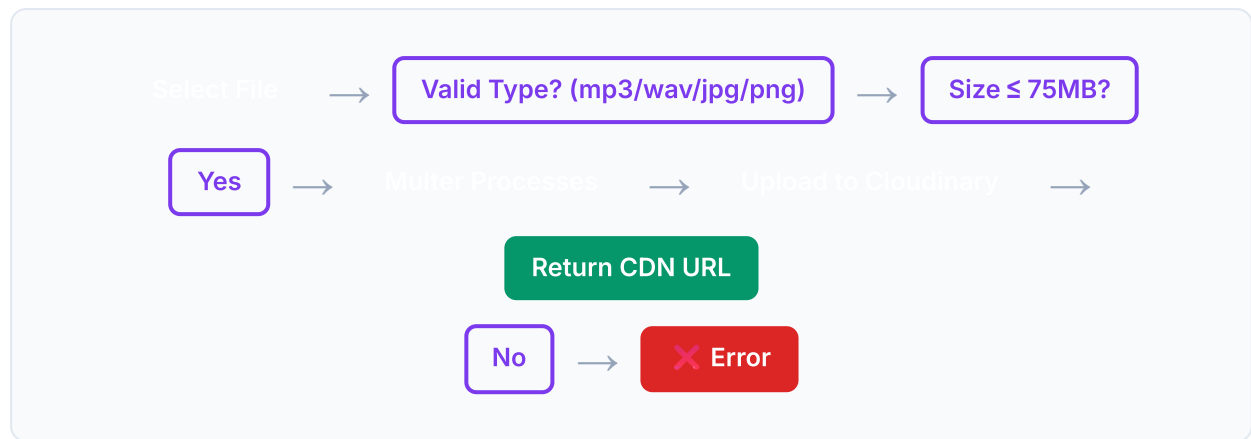
7.2 Admin Authentication Flow



7.3 Content Management Flow (Admin)



7.4 Media Upload Flow



8. Database Schema

Collection	Key Fields	Constraints
User	name, email, password, isAdmin	email unique; password bcrypt-hashed
Song	title, artists, album, duration, coverImage, audioFile, dsps, featured, order	dsps = {spotify, apple, youtube}; max 4 featured
Beat	title, price, audioFile, coverImage, tags, featured	price as display string
Booking	name, email, phone, serviceType, preferredDate, message, status	status: pending confirmed completed cancelled
Contact	name, email, message	timestamps auto-generated
Gallery	title, imagePath, featured	imagePath = Cloudinary URL
Service	title, description, price, iconClass, featured	iconClass defaults to "fas fa-music"

9. API Endpoints

Method	Endpoint	Access	Description
POST	/api/auth/login	Public	Email/password login
POST	/api/auth/google	Public	Google OAuth login
GET	/api/auth/profile	Private	Get user profile
GET	/api/songs	Public	Get all songs
GET	/api/songs/featured	Public	Get featured songs (max 4)
POST	/api/songs	Admin	Create song
PUT	/api/songs/:id	Admin	Update song
DELETE	/api/songs/:id	Admin	Delete song
GET/POST/DEL	/api/beats	Public/Admin	CRUD beats
POST	/api/bookings	Public (Rate Limited)	Create booking
GET/PUT/DEL	/api/bookings/:id	Admin	Manage bookings
POST	/api/contact	Public (Rate Limited)	Submit contact form
GET/POST/DEL	/api/gallery	Public/Admin	CRUD gallery

GET/POST/DEL	/api/services	Public/Admin	CRUD services
POST	/api/upload	Admin	Upload files to Cloudinary (75MB)

10. Security Features

Feature	Implementation	Purpose
Helmet.js	HTTP security headers	Prevents XSS, clickjacking
CORS	Whitelist-based origins	Blocks unauthorized requests
JWT Auth	Bearer tokens, 30-day expiry	Stateless session management
Google OAuth 2.0	Admin email verification	Secure SSO with whitelist
bcryptjs	Salt rounds = 10	One-way password hashing
Rate Limiting	10 req/15 min on forms	Prevents spam
File Validation	Type + size checks (75MB)	Blocks malicious uploads
Env Variables	.env + .gitignore	Secrets never in source control

11. SDG Goals Alignment

This project aligns with the following **United Nations Sustainable Development Goals**:

SDG 4 — Quality Education

Built on open-source technologies (React, Node.js, MongoDB), demonstrating that world-class applications can be created without expensive software. Serves as a practical learning artifact for full-stack development covering MVC architecture, REST APIs, authentication, and cloud deployment. Promotes music production as a creative discipline.

SDG 8 — Decent Work & Economic Growth

Empowers an independent music studio with a professional digital presence. The beat marketplace with pricing enables micro-commerce for digital products. Online booking automates scheduling, reducing overhead. Transparent service listings promote fair business practices and attract more clients.

SDG 9 — Industry, Innovation & Infrastructure

Uses cloud-native architecture (Vercel, Render, Cloudinary, MongoDB Atlas), eliminating physical server needs. Cloudinary CDN delivers media globally. Decoupled frontend/backend with stateless JWT auth enables horizontal scaling. Digitizes traditionally offline studio operations.

SDG 10 — Reduced Inequalities

Free online streaming removes financial barriers to music discovery. Responsive design ensures access from any device. Gives independent artists a professional platform alongside mainstream distribution (Spotify, Apple Music, YouTube). Free-tier hosting makes premium web presence affordable for small studios.

SDG 17 — Partnerships for the Goals

Integrates multiple cloud services (Google OAuth, Cloudinary, MongoDB Atlas, Vercel, Render). Links to Spotify, Apple Music, and YouTube for global distribution. API-first design enables future integrations with payment gateways and third-party services.

12. Conclusion

Rama Records Music Studio is a production-ready MERN stack application demonstrating modern full-stack development. It features a visually appealing frontend with animations and an audio player, a secure RESTful backend with JWT + Google OAuth, cloud-native architecture using Cloudinary/MongoDB Atlas, and a feature-rich admin dashboard. The

project aligns with **five UN SDGs** (4, 8, 9, 10, 17), proving technology can serve both business needs and social objectives.